Beautified Is Not Forged: A Two-Sided Measurement of Face-Forgery Detectors under Beautification and Skin Smoothing

Anonymous submission

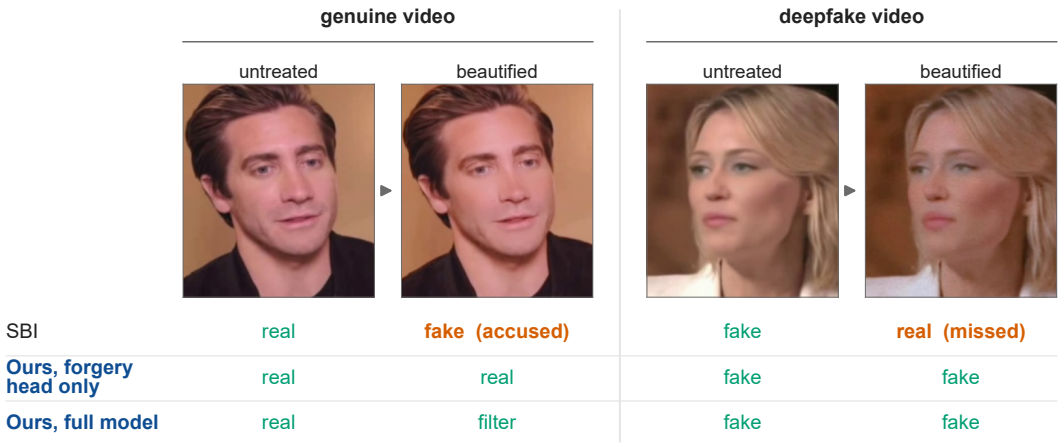


Fig. 1. Celeb-DF-B frames of a genuine video and of a deepfake, before and after the same app preset. SBI, at the threshold that accuses 5% of untreated genuine frames, accuses the beautified genuine face and lets the beautified deepfake pass; our forgery head, at the same rule, accepts the first and catches the second, and our full model calls the first *filter* and the second *fake* (green: correct, orange: error). The frames were selected where SBI errs after beautification and the full model is correct, so they show the two errors, not their rates. Over the benchmark SBI accuses 1.8% of beautified genuine frames and misses 79.8% of beautified deepfakes; the forgery-head-only model accuses 7.4% and misses 36.6%; the full model accuses 9.5%, sends 47.4% to *filter* and misses 40.7% (Table III).

***Abstract*—Many genuine faces reaching a forgery detector were beautified by a camera app, an AR filter or a retouching service. We measure both errors of seventeen published detectors with one rule: the threshold on the forgery probability accuses 5% of each source's untreated genuine faces. All seventeen accuse Instagram AR renders on at least one of two face sets, sixteen on both. On 2,210 FFHQ originals and their commercial renders, fifteen score strong skin smoothing above the same untouched face in 89–99% of pairs, and a threshold fixed on FF++ validation still accuses 15–93% of those faces. Smoothing restricted to the skin is accused by all seventeen; for the four CLIP-based detectors the forgery logit rises with its strength, while a radial vignette alone does not move three of them toward forged. In FF++ training the class that receives beautified views sets the sign of the error: genuine-only smoothing views miss 96–100% of bilaterally smoothed forgeries, forged-only views (one seed) raise the accusation of smoothed genuine faces, and views on both classes sit in between. A smoothing view drawn identically for both classes removes the accusation of whole-crop and commercial smoothing under LoRA tuning of CLIP, not under Effort's tuning scheme and not for skin-only smoothing (38–99% false alarms). Our full-data models accuse 0.1–0.5% of commercially smoothed faces at the FF++-validation threshold but 75–94% of AR renders, so our training does not remove the accusation in general. Evidence maps supervised with exact edit footprints localise held-out part-level forgeries (IoU 0.83).**

I. Introduction

Face-forgery detectors are used for content moderation, remote identity checks and fact-checking. In these settings a genuine face called fake accuses a real person, and a forgery called real is let through. Many genuine faces have been beautified: retouching is common on social media and appears even on identification documents [1], [2], skin smoothing, whitening, eye enlargement and face slimming are default options in camera apps, AR filters and retouching services, and colour presets ship with every photo-sharing app. Beautification edits the pixels a forgery detector inspects but does not change who is shown or what the person does. Biometric work has measured its effect on face recognition [2] and detected retouching as such [1], [3]; how it moves a forgery detector in each direction has been measured for at most six detectors at a time [4]–[6].

We score seventeen published detectors with one rule that counts both errors: Xception [7], EfficientNet-B4 [8], F3Net [9], SPSL [10], UCF [11], RECCE [12], CORE [13], SRM [14], SBI [15], CADDM [16] and LAA-Net [17], the CLIP-based Effort [18], Forensics Adapter [19], GenD [20] and DFD-HR [21], and two general generated-image detectors, UnivFD [22] and NPR [23]. They are scored on four kinds of beautification: app presets, colour filters, Instagram AR

filters [24], and a commercial retouching service where each render can be compared with its own original without a threshold. The largest failure is skin smoothing. The detectors score a smoothed genuine face as more forged than the same face untouched in nearly every pair, smoothing the skin alone with the background untouched is enough, and for the four CLIP detectors the forgery logit of a genuine face grows with the smoothing strength. A radial vignette alone does not produce this shift.

We then ask what training does to these errors, holding architecture, data and recipe fixed and varying only which class receives beautified views. This is shortcut learning [25] and class-dependent augmentation [26] in a setting where the sign of the resulting error can be measured. Views on genuine faces only, the direction of fine-tuning on enhanced genuine samples [5], teach that smoothed means genuine, and a forger who smooths the forgery is let through; views on forged faces only teach the opposite and accuse smoothed genuine faces. A smoothing view drawn in the same way for both classes removes the accusation of whole-crop and commercial smoothing for LoRA-tuned CLIP, at a stated cost in misses on smoothed forgeries; it does not transfer to Effort's tuning scheme or to skin-only smoothing.

Our contributions are:

- **A two-sided measurement with one rule.** Seventeen detectors on four beautification sources, with re-encoding separated from beautification, a JPEG control for the AR renders, a commercial service read pair by pair, a scripted skin-only smoothing test, the dose response of the four CLIP detectors, and a threshold fixed on FF++ validation next to the per-source one.
- **Which class sees beautification.** Four exposure arms (none, genuine only, both, forged only) showing that the class receiving the views sets the sign of the error without changing separability, and a pre-registered chain on skin smoothing with an evasion route, a class-asymmetric form learnt as a label, and a class-symmetric view with its costs and the cases it does not cover.
- **Reference models and a restoration test.** A full-data CLIP model that outputs three classes and four operation scores, and the same model trained with the forgery head only as the detection reference, reported with their weaknesses on two seeds (the shared beautified head costs reenactment; entire-face synthesis is weak; AR renders are accused), and the full model's evidence maps evaluated by localisation and by restoring the named region.

Methods that did not help are summarised in Sec. V-G.

II. Related Work

Beautification and forgery detection. Libourel et al. [4] released Celeb-DF-B and report video AUC drops for three detectors under app presets; Saeed et al. [5] show that face enhancement lets deepfakes evade six detectors and fine-tune on enhanced samples; Concas et al. [6] report both errors for two CNNs under a smoothing filter. None scores detectors by a common rule or trains against the edit with a class-controlled design. On smartphone photographs the false-alarm rate of a generated-image detector is set by where its threshold was calibrated [27]. OpenFilter [24] renders Instagram AR filters onto FairFace and LFW. Retouching detection and restoration [1], [3], [28]–[32] have no forgery class; Liu et al. [33] attribute forgery and beautification tools in a closed set. Perturbation benchmarks for forgery detection [34], [35] and for classification [36] treat edits as corruptions of one class.

Shortcuts and attacks. Shortcut learning [25] and group shifts [37] describe a model that learns a feature correlated with the label; Balestriero et al. [26] show that augmentation and regularisation act differently per class. StatAttack [38] and AdvShadow [39] optimise natural appearance changes so that forgeries pass, Fatima et al. [40] train on the worst of random counter-forensic transforms, and semantic adversarial training exists for classification [41]–[44].

Generalisable forgery detection. Face X-ray [45] and SBI [15] synthesise blending artefacts, UCF [11] separates common from method-specific features, LSDA [46] augments forgery features in latent space for cross-method generalisation, and Effort, Forensics Adapter, GenD and DFD-HR adapt CLIP ViT-L/14. The alpha-blending hypothesis [47] holds that frame-based detectors read compositing artefacts, and its authors ensemble a blending detector with blend-resistant models. DeepfakeBench [48] and DF40 [49] provide checkpoints and test sets.

Explanations. Joint detection and segmentation [50], [51], attention trained so that erasing it removes the evidence [52], amortised maskers [53], [54], DOLOS [55] and FakeVLM [56] precede our footprint-supervised maps; our addition is a restoration test against the true footprint.

III. Problem and Evaluation Rule

Classes. A face image is *forged* if it shows a person, an expression, a motion or speech that was not captured: another identity in whole or in part, a reenactment or lip-sync, or a face or a facial feature (eye, nose, mouth) replaced by new content. It is *beautified* if its colour, tone, skin texture or the smooth shape of the person's own features were edited after capture while identity, expression, pose and speech stay those of the capture. The label follows what changed, not the tool: a retouching service that smooths skin with a neural network beautifies, and a reenactment that keeps the identity forges. The two attributes are independent, and a beautified forgery is forged. Table I lists the cases near the boundary (17 rows in Table S1). A part of the same person regenerated by inpainting was not captured, so we label it forged; we have not measured identity or expression preservation for any edit family, so the boundary is stated, not operationalised.

Requirement and rule. A detector should return a forgery score that does not move when a face is beautified; whether a face was beautified may be reported to the user as a separate attribute (evaluated in Sec. S20), and every detection number reads the forgery score only. Each detector is scored by

TABLE I

CASES NEAR THE CLASS BOUNDARY. "SCORED": WHETHER A TEST SET OF THIS PAPER CONTAINS THE CASE.

Case	Label	Scored
Colour preset, skin smoothing, whitening, slimming, eye enlargement, AR beauty filter (neural or not)	beautified	presets, filters, AR renders, one service
Face swap; reenactment	forged	FF++, Celeb-DF, DFD, DF40
A part from another person	forged	part-level set
A part of the same person regenerated by inpainting	forged (boundary)	part-level set
Beautified forgery	forged	Celeb-DF-B, smoothed forgeries
Entire-face synthesis	forged	DF40, six methods (probe)
Attribute editing, lip-sync	forged	not scored (DF40 editing sets not available to us)

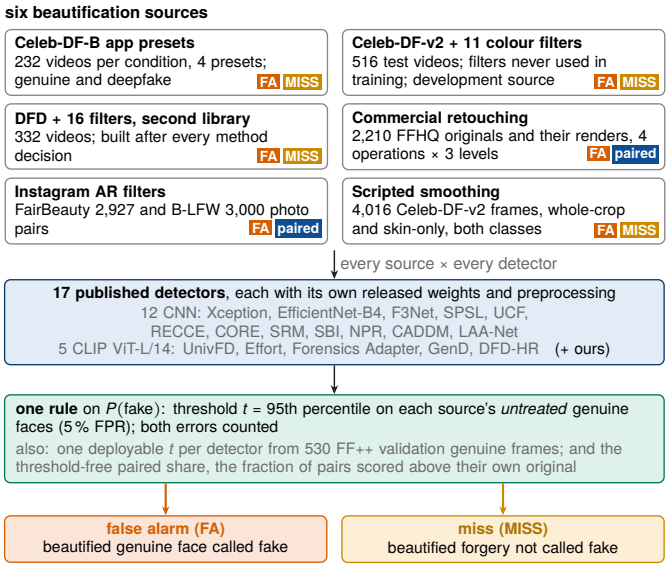


Fig. 2. Measurement protocol. Every detector is scored on every source with its own released weights and preprocessing, by one rule on $P(\text{fake})$. FA: beautified genuine faces called fake; MISS: beautified forgeries not called fake; paired: share of pairs in which the beautified face scores above its own original, which needs no threshold. The Celeb-DF-v2 filter source is a development source.

$P(\text{fake})$ (Fig. 2) with the threshold that accuses 5% of each source's untreated genuine frames. No deployed detector can fit it per source, so we also report one deployable threshold per detector (95th percentile on the 530 genuine FF++ c23 validation frames, 133 videos) and, on the service, statistics that need no threshold. Under beautification we report the false-alarm rate FA (genuine faces called fake) and the miss rate MISS (forgeries not called fake). A forgery that a three-way model sends to its beautified class is a miss. Every AUC is labelled frame or video level; video scores are means of frame scores. Intervals are 95% percentile bootstraps that resample videos (Celeb-DF-B, the filter sources, DF40), originals (the service) or pairs (AR renders), with the threshold re-estimated in every draw: 1,000 draws, 500 for the filter sources. We repeated every interval on the video sources with identities as the resampling unit (Sec. S9); where this changes a reading the text says so. Differences between models use identical draws. We do not correct for multiple comparisons except where stated.

Sources. *Celeb-DF-B* [4]: 232 Celeb-DF-v2 videos per condition (as recorded, re-encoded, beautified by one of four app presets; genuine and synthesised), 32 frames each. The treated videos are re-framed to portrait with tinted bars that hid 57% of their faces from our detector; after reading that, we trimmed uniform bars in all six conditions (post hoc). Face size and resampling still differ between untreated and treated frames, which the re-encoded control does not reproduce. $\Delta\text{MISS}_{\text{beauty}}$ and $\Delta\text{FA}_{\text{beauty}}$ are the rise under the preset minus the rise under re-encoding, and $\text{AUC}_c - \text{AUC}_b$ is the AUC lost beyond re-encoding. *Celeb-DF-v2 filters*: 516 test videos [57], each frame also under one of eleven colour filters [58] never used in training; this set was used to diagnose the filter effect and is a development source. *DFD filters*: 332 DFD videos [59] under one of sixteen filters from a second library, built after every method decision. *AR filters* [24]: FairBeauty, 2,927 FairFace photographs and their Instagram AR renders (eight distinct filters), and B-LFW, 3,000 of its 11,891 LFW pairs (2,016 identities). The AR engine re-renders the face through its own capture path, so we also score FairBeauty with both sides re-compressed (JPEG q90); the AR numbers are an upper bound on the effect of the beautification itself. *Commercial retouching*: 2,210 FFHQ originals and their renders by Alibaba from RetouchingFFHQ [29] (four operations, three levels). No model was trained on it, but more than 60 of our checkpoints and all 17 published detectors were scored on it, Smoothing-90 was singled out after reading it and the bars of Sec. V-D are defined on it, so it is an analysis set, not a clean test. Its originals share no image with our FFHQ training originals (disjoint index range). *Scripted smoothing*: 4,016 Celeb-DF-v2 test frames (1,348 genuine, 513 videos) with a whole-crop bilateral filter at three strengths, a median filter, a Gaussian blend, and a bilateral filter inside the landmark face hull at two strengths (*face-local*, background untouched), applied to both classes. *Untreated detection*: Celeb-DF-v2, DFD and the held-out methods of DF40 [49].

IV. METHOD

Every edited training image has a pixel-aligned original x_0, so the footprint of the edit is exact, $M(x) = \text{dilate}_\rho(\mathbf{1}[\|x - x_0\|_\infty > \tau])$ with $\tau = 6$ on 8-bit intensities.

FF++ triplets. FF++ c23 [60], official 720/140/140 video split. For each genuine training frame x_0 a step draws a genuine item, a beautified item (one of four landmark and skin operations at a random strength, one of 14 colour filters [58], or a random colour grading) and a forged item (a self-blend of x_0 [15], an observed FF++ forgery, or a part-level forgery), and the forged item is beautified with probability 0.5. Forged items without landmarks receive skin smoothing as a whole-crop

TABLE II

MODELS AND ARMS. ALL USE CLIP VIT-L/14 WITH RANK-16 LORA (OR EFFORT'S SVD-RESIDUAL TUNING WHERE NOTED) AND THE TRAINING PIPELINE OF SEC. IV. "VIEWS": WHICH CLASS RECEIVES BEAUTIFIED TRAINING VIEWS.

Name	Data, views, head	seeds
P0, P1, P2	FF++ only, forged head; views on no class, genuine only, both	3
P3	as P0, views on forged faces only	1
P1-smooth	P1 with smoothing as the only view	3
P2-smooth, P2-nosmooth	P2 with smoothing only, with smoothing removed	2
P2-sym	smoothing as the only view, same form on both classes	2
Effort scheme	P0, P2 (2 seeds), P2-sym (1), SVD-residual tuning	1–2
OURS, forgery head only	full data†, P2 views, forgery head only (binary)	2
OURS, full model	same; three classes + operations, evidence map	2
OURS, full model, sym. view	OURS, full model with the class-symmetric smoothing view	2

†FF++ plus 7,831 FFHQ originals, their renders by two retouching services (Tencent, Megvii) and the single-operation components of those renders, and 61,510 part-level forgeries. The published CLIP detectors were trained on FF++ alone.

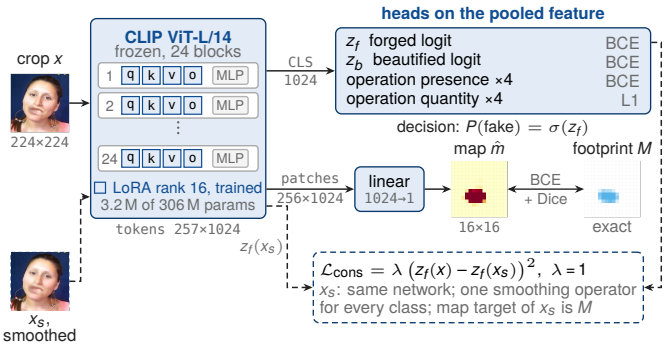


Fig. 3. The full model. A frozen CLIP ViT-L/14 with rank-16 LoRA on the q, k, v and output projections of all 24 blocks; linear heads on the pooled feature give the forged logit z_f, the beautified logit z_b and four operation presence and quantity scores; a linear layer on the 256 patch tokens gives the 16×16 evidence map $\hat{m}$, trained against the exact footprint M of the edit. The forgery decision reads $P(\text{fake}) = \sigma(z_f)$ only; OURS, forgery head only keeps z_f alone. Dashed: a pre-registered training-time variant that is not part of any model reported here, in which a skin-smoothed copy x_s of every item passes through the same network and $\mathcal{L}_{\text{cons}}$ ties $z_f(x_s)$ to $z_f(x)$. Example: a held-out part-level forgery (mouth of another identity) and the map of the full model.

filter, not as the face-local operation; Sec. V-D shows what this asymmetry teaches. The same random blur or downscale is applied to all three items. The FF++-only arms use nothing else.

Commercial renders and part-level forgeries. The full-data models add 7,831 FFHQ photographs [61] and their renders by Tencent and Megvii [29], sister services of the test service with the same four operations. Each render is split by dense optical flow [62] and a photometric residual into eye and contour warps and low- and high-frequency photometric components in the original's geometry (supplement, Alg. S1); 50,112 of 73,952 components lie above the noise floor. A render's photometric residual is also added to a self-blend of the same original. 61,510 part-level forgeries on FF++ frames transplant the eyes, nose or mouth of another identity or regenerate the part with Stable Diffusion inpainting, with pixels outside the feathered mask copied from x_0; the held-out set uses test frames and test donors.

Model (Fig. 3). A frozen CLIP ViT-L/14 [63] with rank-16 LoRA [64] on every attention projection (3.2 M of 306 M parameters trained). In the full model two linear heads on the pooled feature give the forged logit z_f and the beautified logit z_b; $P(\text{fake}) = \sigma(z_f)$ and $P(\text{filter}) = (1 - \sigma_f)\sigma_b$. An operation-presence head and a 1×1 convolution on the spatial tokens (the evidence map $\hat{m}$, 16×16) are trained against the operation labels and M with BCE and Dice; the classifier does not read $\hat{m}$. The system outputs three classes and four operation scores; the forgery decision is read from z_f alone, and the beautified class and operation scores are attributes reported separately. The forgery-head-only model has the same data, views and backbone and only z_f; it shows what the forged head reaches without the shared beautified head, and it is the first row of every detection table because it is better than or equal to the full model on every untreated source and on Celeb-DF-B on both seeds while the shared head costs reenactment (Sec. V-E; ordering chosen after both seeds were read). Training: 224 px, AdamW at 10^{-4}, 8 epochs, 16 genuine bases per batch. Checkpoints are chosen on validation data only (FF++ validation AUC with the forgery head only; for the full model its macro-F1 averaged with balanced accuracy on held-out originals and renders of the training services).

V. RESULTS

A. *Both errors under one rule*

Commercial retouching. Table IV reads the Smoothing-90 renders against their own originals. Fifteen of the seventeen detectors score the render above its original in 89–99% of pairs, and the median $P(\text{fake})$ moves from 0.21 to 0.99 for SBI and from 0.23 to 0.92 for GenD: a heavily smoothed genuine face is scored like a forgery whatever the threshold. The exceptions are NPR, saturated, and CADDM (60.9%). At the threshold that accuses 5% of the originals thirteen detectors accuse 36.3–68.7% of the renders. Whitening, face slimming and eye enlarging at their strongest level stay at 3.7–8.1% for thirteen of the sixteen detectors other than NPR (Table S29); these operations go through the same service, encoder and resolution as the smoothing, so the retouching result is a smoothing result. A sign test rejects "no shift" for every published detector other than NPR ($p < 10^{-10}$), with median paired logit shifts of $+0.7$ to $+5.5$. Our full-data models accuse 6.5–12.5% of the renders, but none leaves the score unmoved: OURS, forgery head only still ranks the render above its original in 70.5 and 68.2% of pairs (median $+0.41$ [$+0.38$, $+0.44$] and $+0.19$), and OURS, full model moves it the other way, below its original in 67.5% of pairs (median -1.20 [-1.28, -1.12]; second seed -0.53). Only the FF++-only symmetric arm P2-sym stays close to no shift ($+0.09$ [$+0.03$, $+0.12$] and $+0.03$ [0.00, $+0.09$]). Our full-data models were

TABLE III

BOTH ERRORS UNDER ONE RULE ($P(\text{fake})$, THRESHOLD AT 5% FPR ON EACH SOURCE'S UNTREATED GENUINE FRAMES). CELEB-DF-B: AUC UNTREATED, RE-ENCODED (AUC_c) AND BEAUTIFIED (AUC_b), AUC LOST BEYOND RE-ENCODING, RISE IN MISSES AND IN FALSE ALARMS UNDER THE PRESET MINUS THE RISE UNDER RE-ENCODING, AND FA_b, THE FALSE-ALARM RATE ON BEAUTIFIED GENUINE FRAMES. DFD: SIXTEEN HELD-OUT FILTERS ON THE FULL FRAME. PERCENTAGE POINTS. INTERVALS, THE CELEB-DF-V2 FILTER SOURCE AND VIDEO-LEVEL AUC IN THE SUPPLEMENT (TABLES S4 AND S5). ‡GENERAL GENERATED-IMAGE DETECTORS, NOT TRAINED ON FACES. †TRAINED ON MORE DATA THAN THE PUBLISHED DETECTORS (TABLE II).

		Celeb-DF-B, app presets (frame level)							DFD + 16 held-out filters (frame level)			
Detector	train	AUC	AUC_c	AUC_b	$\text{AUC}_c - \text{AUC}_b$	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	$\Delta\text{FA}_{\text{beauty}}$	AUC	AUC_f	ΔMISS	ΔFA
Xception [7]	FF++ c23	0.811	0.795	0.737	0.058	+10.9	5.7	-1.7	0.854	0.834	+2.0	+2.0
EfficientNet-B4 [8]	FF++ c23	0.811	0.798	0.762	0.035	+3.4	7.8	+1.7	0.810	0.781	+6.3	-0.5
F3Net [9]	FF++ c23	0.798	0.782	0.731	0.051	+10.9	5.4	-1.9	0.781	0.774	-8.5	+3.6
SPSL [10]	FF++ c23	0.781	0.769	0.727	0.042	+12.4	3.4	-2.4	0.829	0.837	+4.7	-1.8
UCF [11]	FF++ c23	0.841	0.829	0.777	0.053	+11.3	5.7	+0.2	0.822	0.804	+5.0	+0.3
RECCE [12]	FF++ c23	0.785	0.765	0.723	0.042	+13.8	4.1	-3.6	0.833	0.808	+7.3	-0.2
CORE [13]	FF++ c23	0.821	0.808	0.752	0.057	+13.5	4.0	-2.2	0.827	0.821	-5.4	+1.6
SRM [14]	FF++ c23	0.789	0.769	0.733	0.035	+15.3	2.8	-3.4	0.830	0.822	+5.8	+0.2
SBI [15]	FF++ c23	0.875	0.861	0.752	0.109	+21.8	1.8	-0.6	0.829	0.810	+4.5	+0.4
UnivFD‡ [22]	ProGAN‡	0.653	0.637	0.624	0.013	-7.2	7.6	+4.5	0.840	0.801	-10.2	+11.6
NPR‡ [23]	ProGAN‡	0.503	0.503	0.498	0.006	-0.7	3.4	+0.9	0.473	0.516	-13.7	+9.4
Effort [18]	FF++ c23	0.909	0.882	0.807	0.075	+11.0	5.4	+2.1	0.939	0.905	+15.1	-2.3
Forensics Adapter [19]	FF++ c23	0.909	0.894	0.828	0.066	+15.2	7.9	+0.1	0.940	0.924	+8.0	-0.6
GenD [20]	FF++ c23	0.924	0.898	0.807	0.091	+12.0	9.9	+3.6	0.934	0.925	+3.1	+1.2
DFD-HR [21]	FF++ c23	0.938	0.915	0.838	0.077	+3.6	14.6	+9.8	0.943	0.928	+9.1	-2.0
CADDM [16]	FF++ raw	0.886	0.850	0.762	0.088	-2.9	13.8	+9.9	0.845	0.818	+3.8	+0.8
LAA-Net [17]	FF++ raw	0.889	0.870	0.698	0.172	+14.7	6.0	+3.3	0.810	0.783	+6.2	+0.0
P0	FF++	0.927	0.901	0.860	0.041	+17.0	5.1	-1.6	0.943	0.924	+9.3	-0.2
P2-sym	FF++	0.920	0.900	0.866	0.034	+14.2	5.5	-1.2	0.936	0.921	+7.3	-0.4
OURS, forgery head only†	FF++, FFHQ	0.949	0.934	0.896	0.038	+7.0	7.4	+2.3	0.961	0.952	+4.2	+0.2
second seed†	FF++, FFHQ	0.955	0.939	0.904	0.035	+5.7	8.3	+3.2	0.951	0.938	+3.8	+1.1
OURS, full model†	FF++, FFHQ	0.938	0.918	0.866	0.052	+6.7	9.5	+4.9	0.942	0.931	+8.9	-1.0
second seed†	FF++, FFHQ	0.941	0.926	0.883	0.043	+8.3	6.7	+2.7	0.945	0.931	+4.4	+0.3
OURS, full model, sym. view†	FF++, FFHQ	0.941	0.925	0.868	0.057	+10.0	9.2	+4.9	0.947	0.934	+8.5	-0.9
second seed†	FF++, FFHQ	0.945	0.934	0.888	0.046	+3.1	8.7	+4.9	0.954	0.940	+2.6	+2.9

TABLE IV

COMMERCIAL SKIN SMOOTHING (SMOOTHING-90): 2,210 FFHQ ORIGINALS AND THEIR RENDERS. SHARE OF RENDERS SCORED ABOVE THEIR OWN ORIGINAL (50 MEANS NO SHIFT), MEDIANS OF $P(\text{FAKE})$, MEDIAN PAIRED SHIFT OF THE FORGERY LOGIT WITH A BOOTSTRAP INTERVAL OVER ORIGINALS, SHARE OF RENDERS ABOVE THE THRESHOLD THAT ACCUSES 5% OF THE ORIGINALS, AND SHARE ABOVE THE THRESHOLD FIXED ON FF++ VALIDATION. NPR SCORES EVERYTHING AT 1.0.

Detector	above own orig. (%)	median $P(\text{fake})$	median Δlogit [95% CI]	accused (%)	FF++-val. thr. (%)
Xception [7]	95.2	0.29 → 0.94	+3.32 [+3.17, +3.45]	51.6	65.0
EfficientNet-B4 [8]	93.7	0.32 → 0.96	+3.48 [+3.33, +3.57]	36.3	63.8
F3Net [9]	97.8	0.13 → 0.97	+4.92 [+4.74, +5.08]	50.6	69.0
SPSL [10]	97.3	0.46 → 0.96	+3.05 [+2.95, +3.12]	51.2	56.3
UCF [11]	95.4	0.03 → 0.98	+5.47 [+5.20, +5.73]	41.4	48.4
RECCE [12]	96.5	0.12 → 0.94	+3.82 [+3.70, +3.96]	36.8	60.0
CORE [13]	96.6	0.25 → 0.95	+3.90 [+3.74, +4.03]	54.8	66.7
SRM [14]	99.3	0.46 → 0.67	+0.68 [+0.65, +0.71]	62.8	62.0
SBI [15]	96.2	0.21 → 0.99	+4.88 [+4.72, +5.04]	37.3	93.3
UnivFD‡ [22]	90.4	0.01 → 0.02	+1.22 [+1.17, +1.28]	16.6	15.2
NPR‡ [23]	n/a	n/a	n/a	n/a	n/a
Effort [18]	95.6	0.32 → 0.59	+0.96 [+0.90, +1.02]	43.2	31.1
Forensics Adapter [19]	93.3	0.06 → 0.35	+1.72 [+1.59, +1.87]	39.7	32.5
GenD [20]	98.5	0.23 → 0.92	+3.27 [+3.19, +3.35]	68.7	67.4
DFD-HR [21]	99.0	0.21 → 0.71	+2.06 [+1.93, +2.15]	64.9	56.2
CADDM [16]	60.9	0.54 → 0.74	+0.67 [+0.53, +0.81]	6.3	42.3
LAA-Net [17]	89.2	0.23 → 0.97	+3.60 [+3.36, +3.79]	17.2	81.0
P0	98.5	0.05 → 0.97	+5.17 [+5.05, +5.38]	54.2	–
P2-sym	53.7	0.02 → 0.02	+0.09 [+0.03, +0.12]	5.9	–
OURS, forgery head only†	70.5	0.00 → 0.00	+0.41 [+0.38, +0.44]	9.1	0.2
second seed†	68.2	0.00 → 0.00	+0.19 [+0.16, +0.19]	6.5	0.1
OURS, full model†	32.5	0.00 → 0.00	-1.20 [-1.28, -1.12]	8.6	0.5
OURS, full model, sym. view†	30.0	0.00 → 0.00	-1.09 [-1.19, -1.00]	8.1	0.1

trained on renders of the two sister services (marked †), which the published detectors were not.

A deployable threshold. With the threshold fixed on FF++ validation (last column), the published detectors accuse 15.2% (UnivFD) to 93.3% (SBI) of the Smoothing-90 renders, and CADDM and LAA-Net, which accuse 6.3 and 17.2% under the per-source rule, accuse 42.3 and 81.0%: their scores drift on untreated stills, and SBI, CADDM and LAA-Net already call 45.3, 32.9 and 30.9% of the untouched FFHQ originals fake. Our models accuse 0.1–0.5% (both seeds); they were trained on FFHQ stills and sister-service renders, so this is partly "trained on stills of this kind". The detector ranking by accused share is not stable under the change of rule (Spearman between the two rules 0.29, $p = 0.25$, on the service; 0.37, $p = 0.15$, on FairBeauty), so the threshold-free columns are the ones to read first (Table S9).

Instagram AR filters. Under the per-source rule all seventeen detectors accuse FairBeauty renders above the 5% floor (lower interval bound above 5%; 10.7–85.9%) and sixteen accuse B-LFW renders (12.0–96.6%); NPR's scores saturate on the 112 px B-LFW renders. Every published detector ranks the FairBeauty render above its own original in 68.9–98.8% of pairs, and all but NPR rank the B-LFW render above it in 80.7–99.9% (Fig. 4, Table S8). With both sides of FairBeauty re-compressed, every detector except NPR keeps 0.64–1.05 of its accused share, so re-compression does not carry the effect, although the AR capture path remains confounded with the beautification. Which detectors accuse most depends on the source (Spearman of the FairBeauty and service shares over the seventeen 0.15, $p = 0.58$); that every detector accuses does not. Our models are not exempt: OURS, forgery head only accuses 79.0–82.1% of FairBeauty and 89.0–90.3% of B-LFW renders (two seeds), more than Effort and Forensics Adapter (first seed, paired $+11.3$ [$+7.6$, $+15.1$] and $+15.1$ [$+11.8$, $+18.1$] on FairBeauty) and less than DFD-HR; OURS,

	Presets CDF-B	Colour DFD	Service S-90	AR FB	AR B-LFW	Face-local
Xception	6	7	52	24	53	89
EfficientNet-B4	8	5	36	22	40	64
F3Net	5	9	51	26	49	86
SPSL	3	3	51	11	12	70
UCF	6	5	41	37	71	89
RECCE	4	5	37	41	78	81
CORE	4	7	55	25	59	85
SRM	3	5	63	27	58	81
SBI	2	5	37	23	35	97
CADDM	14	6	6	15	36	88
LAA-Net	6	5	17	28	48	100
UnivFD	8	17	17	63	49	14
NPR	3	14	n/a	70	0	68
Effort	5	3	43	68	75	88
Forensics Adapter	8	4	40	64	77	96
GenD	10	6	69	76	97	96
DFD-HR	15	3	65	86	95	99
Ours, forgery head only	7	5	9	79	89	52
Ours, full model	10	4	9	75	87	6
Full model, sym. view	9	4	8	77	94	38

Fig. 4. Accused share of beautified genuine faces (%) at the threshold that accuses 5% of each source's untreated genuine faces: Celeb-DF-B presets (FA_b), DFD under sixteen colour filters, commercial Smoothing-90, Instagram AR renders of FairBeauty (FB) and B-LFW, and the scripted face-local smoothing of Celeb-DF-v2 genuine frames (strength 0.06). Rows: eleven face-forgery CNNs, two general detectors, four CLIP detectors, and our full-data models (first seed). NPR is saturated on the service.

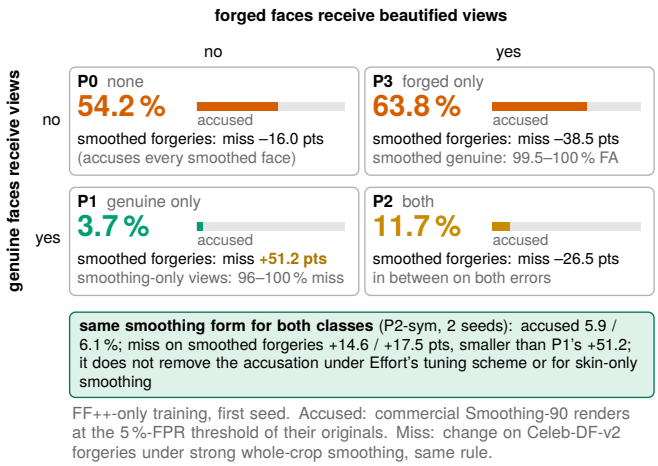


Fig. 5. Which class receives beautified training views decides the sign of the error (FF++-only arms, first seed; Table V). Accused: commercial Smoothing-90 renders at the threshold that accuses 5% of their originals. Miss: change of the miss rate on Celeb-DF-v2 forgeries under strong whole-crop smoothing at the same rule; a negative change comes from accusing every smoothed face. P3 has one seed.

full model accuses 75.1 and 86.7%. At the FF++-validation threshold OURS, forgery head only calls 6.5–13.3% of the untouched FairFace originals fake (OURS, full model 19.1%). Our training removes the accusation on the vendor family it saw, not on AR filters.

Skin-only smoothing. A bilateral filter inside the face hull, with the background untouched, is accused by all seventeen detectors (Fig. 4, Table S10): the four CLIP detectors call 87.8–99.3% of genuine frames forged at the milder strength and 99.9–100% at the stronger one, the eleven face-forgery CNNs 64.3–100% and 91.8–100%, UnivFD 13.9 and 55.2%, NPR 68.0 and 77.7%. At matched visible strength skin-only smoothing is accused more often than whole-crop smoothing.

App presets. On Celeb-DF-B every detector that detects the untreated forgeries loses frame AUC beyond what re-encoding costs (Table III): SBI 0.861 to 0.752 against the re-encoded frames, CADDM 0.850 to 0.762, LAA-Net 0.870 to 0.698, and 0.066–0.091 for the four CLIP detectors. At video level CADDM goes from 0.965 to 0.832 (0.933 re-encoded), against 0.91 to 0.76 reported by Libourel et al. [4]. Beyond re-encoding the presets raise the miss rate by 21.8 (SBI), 11.0 (Effort), 15.2 (Forensics Adapter), 12.0 (GenD) and 14.7 points (LAA-Net), and false alarms by 9.8 (DFD-HR), 9.9 (CADDM) and 3.6 points (GenD). CADDM's misses do not rise (-2.9; under identity resampling its interval contains zero) because its genuine scores rise too. Under the DFD filters Effort's miss rate rises by 15.1 points and Forensics Adapter's by 8.0.

B. Dose response

On the 1,348 genuine Celeb-DF-v2 frames we varied the strength of three edits and read the forgery logit of the four released CLIP detectors (Table S11). Under face-local smoothing the logit rises in at least 97.9% of frames at every level for all four and the median shift grows with strength; the weakest level (0.03) already moves the median logit by $+2.396$ (Effort) to $+4.585$ (Forensics Adapter), 1.0–1.9 times the gap between the mean forged and the mean genuine logit of the detector, and the strongest by 1.9–3.0 gaps. Whole-crop bilateral smoothing gives the same picture at smaller shifts (1.0–1.7 gaps at the strongest strength). A radial vignette alone does not: Effort, Forensics Adapter and GenD move the other way (-0.2 to -0.4 at strength 0.8), and DFD-HR passes the monotonicity rule with a shift of 3% of its gap. For the published CLIP detectors smoothing is the operative edit, and a genuine face is pushed past the forged mean. The radial-overlay effect we saw earlier in one FF++-only model of ours (Sec. S13) is a property of that model, not of the released detectors.

C. Which class receives beautified views

With architecture, data and recipe fixed, P0–P3 differ only in which class receives beautified views (Table II, Fig. 5). On Celeb-DF-B the beautification-specific miss rise is 17.0, 14.2 and 10.8 points for P0 (three seeds), 21.5, 21.8 and 20.4 for P1 and 9.8, 10.3 and 12.8 for P2 (Table S13). P1 exceeds P0 by 4.5–9.5 points in every seed and P2 by 7.5–11.7 (Holm-adjusted $p \leq 0.01$ for all three P1$-$P0 contrasts). P2 against P0 is -7.2, -4.0 and $+2.0$ points, changing sign within the seed spread of P0. On the two filter sources P2 lowers the miss rise by 5–10 points in 9 of 10 seed-by-source cells and raises

the false-alarm rise by 2–5 points; the frame AUC under the filters does not change.

The forged-only arm P3 (one seed, pre-declared reading) mirrors P1 on every source (Table S15). It accuses 63.756% [59.050, 67.784] of the Smoothing-90 renders ($+9.6$ [$+6.0$, $+13.4$] over P0), 99.5 and 100% of mid and strong whole-crop-smoothed genuine frames (P0: 16.8 and 39.6%) and 98.1–100% of skin-only-smoothed ones. Smoothed, filtered or beautified forgeries are caught more often than untreated ones (miss change -25.1 points under the Celeb-DF-v2 filters and -6.6 under the DFD filters), and genuine faces pay for it (47.2% of filtered Celeb-DF-v2 genuine frames and 24.6% of beautified Celeb-DF-B genuine frames accused, against 18.8 and 5.1% for P0). The sign rule holds: genuine-only views lower the accusation and open evasion, forged-only views raise the accusation and close evasion, and views on both classes sit in between. Exposure moves errors between misses and false alarms; it does not make the classes more separable (P3 untreated Celeb-DF-v2 video AUC 0.943 against 0.938 for P0). Under Effort's tuning scheme P2 also lowers the miss rise (-4.6 and -5.2 points). P0 already keeps more frame AUC under the presets (0.860–0.883) than released Effort (0.807), so most of the margin of our models over the CLIP baselines comes from the base recipe, not from exposure.

D. Skin smoothing: a chain of pre-registered arms

FF++-only arms, each with bars written before training and no service image (Table V, Sec. S14); the scripted smoothing range was set while the test service's measurements were on disk, and the first service analysis was post hoc.

No exposure reproduces the published detectors. P0 accuses 54.2, 56.1 and 43.5% of the Smoothing-90 renders and scores them above their original in 97–99% of pairs, although it trains with label-symmetric blur, rescaling and JPEG. General beautified views lower the rate to 3.7–7.1% (P1) and 6.9–11.7% (P2), but P2 still ranks the render above its original in 72–84% of pairs. P2-nosmooth accuses 55.3% [51.9, 58.6] and 67.1% (two seeds), so no other view substitutes for smoothing, within the views we trained.

The genuine-only recipe is an evasion route. P1-smooth accuses 0.7–1.0% (three seeds) and ranks the render above its original in 2.6–6.5% of pairs. A check written before any smoothed forgery was scored applies smoothing forms no arm has seen to Celeb-DF-v2 test frames. P1-smooth then misses 96–100% of bilaterally smoothed forgeries (miss rise of 56–63 points), 75–84% of median-filtered forgeries and 51–54% of blended ones, against 37–44% untreated, and its miss rate under skin-only smoothing rises by 41.5 and 55.1 points. It has learnt that smoothed means genuine.

A class-asymmetric form is learnt as a label. A whole-crop bilateral filter on a *genuine* frame is called forged by Ours, full model in 34.9, 79.8 and 90.1% of frames at the three strengths (second seed 22.0, 84.1 and 96.1%) and by P2-smooth in 47.8–88.3%, against 0.8, 16.8 and 39.6% for P0; Ours, forgery head only, trained on the same items, is at 9.9, 35.5 and 45.6% (2.2, 9.3 and 17.2%). In training, genuine items receive the face-local smoothing and forged items without landmarks the whole-crop filter, so the form of the smoothing predicts the class. Ours, full model is the only model whose skin-only false alarms (5.8 and 44.5%) are below its whole-crop ones; at the milder skin-only strength its miss rate rises by 17.6 points.

A class-symmetric view. P2-sym draws both forms on both classes. Bars: (a) Smoothing-90 accused at most 15% with the render above its original in 25–75% of pairs; (b) at most 15% false alarms on smoothed genuine frames for every form; (c) a miss rise at most 5 points above P0's for every form. (a) and (b) are met on both seeds (5.9 and 6.1%, 53.7 and 51.8% of pairs, false alarms 2.2–6.1%). Bar (c) is not met: the miss rises are $+7.6$ to $+20.6$ and exceed P0's by more than 5 points at the mid and strong strengths, partly because P0's miss rate falls under strong smoothing as it accuses everything smoothed. Its frame AUC with both classes smoothed (0.767 and 0.796 at the strong strength) is at or above P0's (0.736–0.747). Two limits were found after the bars were read. Skin-only smoothing, a form the bars did not cover, is still accused: 54.1 and 37.8% at the milder strength, 98.7 and 91.5% at the stronger. Under Effort's tuning scheme the same view meets bar (b) but not (a): Smoothing-90 falls from 56.9 to 14.5%, but the render is still ranked above its original in 88.4% of pairs. The recipe holds for LoRA tuning of CLIP and for the forms it was drawn from.

The full model with the symmetric view, two seeds. The variant replaces Ours, full model only if five pre-declared criteria hold (Table S22). It fails on both seeds, for different reasons: on the first the explanation flip rate falls by 1.4–1.6 points against a bar of 1.0 (the ceiling fell as much), on the second Smoothing-90 is accused at 17.8% against a bar of 15% (Ours, full model 12.5%). The intended effect replicates: false alarms on whole-crop-smoothed genuine frames of 1.0–1.7 and 3.8–8.4% (Ours, full model: 22.0–96.1%), with untreated video AUC kept. The size of its cost does not: the miss rate on strongly smoothed forgeries rises by 34.1 points on the first seed and 11.6 on the second (frame AUC with both classes smoothed 0.796 and 0.784 against 0.688 and 0.711). Its skin-only false alarms are 37.9 and 80.2% (first seed). By the rule Ours, full model is kept.

E. Untreated detection and DF40

Celeb-DF-v2 and DFD. Each DeepfakeBench-trained baseline is scored on its own crop and our models on ours (Table S3). Ours, forgery head only reaches a frame AUC of 0.914 on Celeb-DF-v2 and 0.961 on DFD (second seed 0.923 and 0.951): above Effort on both sets and above all four CLIP detectors on DFD, on par with Forensics Adapter and GenD on Celeb-DF-v2 (paired intervals, first seed). Ours, full model (0.891 and 0.942) is on par with all four; our Effort reproduction is below its published Celeb-DF-v2 video AUC (0.927 against 0.956). Under the presets Ours, forgery head only keeps a frame AUC_b of 0.896 and 0.904 and Ours, full model 0.866 and 0.883; against the four CLIP detectors the paired differences are $+0.058$ to $+0.097$ for Ours, forgery

TABLE V

WHAT EACH TRAINING RECIPE BUYS AND COSTS, PER SEED (1 / 2 / 3 WHERE TRAINED; "–" NOT SCORED); NAMES AS IN TABLE II. COMMERCIAL SMOOTHING-90 AS IN TABLE IV. SMOOTHED CELEB-DF-V2: CHANGE OF THE MISS RATE ON FORGERIES UNDER THE STRONG WHOLE-CROP BILATERAL FILTER AT THE THRESHOLD OF 5% FPR ON UNTREATED GENUINE FRAMES (A NEGATIVE VALUE COMES FROM ACCUSING EVERY SMOOTHED FRAME, GENUINE ONES INCLUDED), AND FRAME AUC WITH BOTH CLASSES SMOOTHED. UNTREATED CELEB-DF-V2 VIDEO AUC; CELEB-DF-B FRAME AUC UNDER THE PRESETS; DF40 REENACTMENT (FR) IN THE FF++ DOMAIN, VIDEO AUC, MEAN OVER TWELVE METHODS. DFD, $\Delta\text{MISS}_{\text{BEAUTY}}$, THE CELEB-DF DOMAIN, EVERY SMOOTHING FORM AND INTERVALS IN TABLES S20, S21, S10 AND S23.

	commercial Smoothing-90		smoothed Celeb-DF-v2		Celeb-DF-v2	Celeb-DF-B	DF40 FR
Arm	accused (%)	above own orig. (%)	miss rise	frame AUC	video AUC	frame AUC_b	video AUC
P0	54.2 / 56.1 / 43.5	98.5 / 99.0 / 97.0	-16.0 / -11.8 / +4.3	0.747 / 0.736 / 0.736	0.938 / 0.938 / 0.940	0.860 / 0.883 / 0.883	0.906 / 0.877 / 0.873
P1	3.7 / 7.1 / 4.6	43.5 / 67.5 / 64.3	+51.2 / – / –	0.587 / – / –	0.900 / 0.910 / 0.923	0.844 / 0.851 / 0.864	0.807 / – / –
P2	11.7 / 7.1 / 6.9	84.1 / 72.4 / 75.5	-26.5 / -13.3 / –	0.740 / 0.746 / –	0.917 / 0.920 / 0.935	0.879 / 0.893 / 0.877	0.809 / 0.785 / –
P3	63.8	99.2	-38.5	0.743	0.943	0.857	0.870
P2-nosmooth	55.3 / 67.1	98.6 / 99.4	-28.9 / -33.1	0.732 / 0.695	0.913 / 0.924	0.879 / 0.873	0.858 / –
P2-smooth	2.3 / 5.3	27.3 / 51.7	-48.2 / -44.6	0.785 / 0.734	0.904 / 0.926	0.865 / 0.870	0.779 / 0.833
P1-smooth	0.7 / 1.0 / 0.9	2.6 / 6.5 / 5.4	+56.4 / +60.0 / +62.9	0.643 / 0.579 / 0.596	0.922 / 0.929 / 0.939	0.887 / 0.896 / 0.893	0.871 / 0.869 / 0.900
P2-sym	5.9 / 6.1	53.7 / 51.8	+14.6 / +17.5	0.767 / 0.796	0.919 / 0.934	0.866 / 0.888	0.871 / 0.796
Effort scheme, P0	56.9 / 63.2	99.1 / 99.1	-32.2 / –	0.719 / –	0.944 / 0.954	0.882 / 0.886	– / –
Effort scheme, P2-sym	14.5	88.4	+24.9	0.757	0.935	0.890	0.822
released Effort fine-tuned, no views	55.3	98.5	-39.9	0.754	0.926	0.805	–
released Effort fine-tuned, P2-sym view	39.9	96.6	-19.0	0.740	0.921	0.791	0.805
OURS, forgery head only	9.1 / 6.5	70.5 / 68.2	-15.0 / +2.2	0.754 / 0.810	0.956 / 0.963	0.896 / 0.904	0.842 / 0.856
OURS, full model	8.6 / 12.5	32.5 / 41.3	-44.3 / -46.7	0.688 / 0.711	0.947 / 0.946	0.866 / 0.883	0.718 / 0.780
OURS, full model, sym. view	8.1 / 17.8	30.0 / 46.2	+34.1 / +11.6	0.796 / 0.784	0.954 / 0.953	0.868 / 0.888	0.738 / 0.718

TABLE VI

DF40 HELD-OUT, UNTREATED: VIDEO AUC, MEAN OVER NINE FACE-SWAP (FS) AND TWELVE REENACTMENT (FR) METHODS, AGAINST THE SAME-DOMAIN REAL VIDEOS. ALL DETECTORS SHARE ONE CROP. INTERVALS AND PAIRED DIFFERENCES IN TABLE S23; ENTIRE-FACE SYNTHESIS IN TABLE S24.

	FF++ domain		Celeb-DF domain	
Detector	FS	FR	FS	FR
eight CNNs (range)	0.766–0.860	0.754–0.859	0.656–0.736	0.671–0.740
Effort [18]	0.944	0.808	0.935	0.710
Forensics Adapter [19]	0.948	0.828	0.897	0.665
GenD [20]	0.966	0.935	0.947	0.828
DFD-HR [21]	0.982	0.932	0.940	0.698
CADDM [16]	0.869	0.794	0.788	0.541
OURS, forgery head only	0.969	0.842	0.926	0.647
second seed	0.961	0.856	0.924	0.698
OURS, full model	0.952	0.718	0.905	0.502
second seed	0.960	0.780	0.928	0.559
OURS, full model, sym. view	0.952	0.738	0.909	0.539
second seed	0.958	0.718	0.932	0.504
P0	0.981	0.906	0.925	0.692
P2-sym	0.968	0.871	0.913	0.670

head only and $+0.028$ to $+0.076$ for OURS, full model over both seeds, every interval above zero under video and identity resampling (Table S6). Our models use more training data than the baselines; the equal-data comparison is P0, 0.022–0.053 above the four, also with every interval above zero. Identity resampling changes the category of 24 of 563 intervals, none of them a headline: six ours-minus-published AUC differences under filters, whose video intervals excluded zero by at most 0.005, and eleven DF40 FF++-domain differences of 0.009–0.034 now contain zero and are ties (Table S12).

Forgery head only against the full model. Trained on the same data, OURS, forgery head only is better than OURS, full model on Celeb-DF-v2 frames ($+0.023$ and $+0.036$), on Celeb-DF-B under the presets ($+0.030$ and $+0.021$) and on DF40 reenactment ($+0.124$ and $+0.076$ in the FF++ domain, $+0.145$ and $+0.139$ in the Celeb-DF domain), every interval above zero on both seeds; on DFD and Celeb-DF-v2 video AUC its advantage holds on one seed, and no interval is below zero. The shared beautified head costs detection, most on reenactment.

DF40. DF40 [49] adds nine face-swap and twelve reenactment methods in an FF++ and a Celeb-DF domain; the list and statistics were fixed before any detector was run on it. On face swap OURS, forgery head only is on par with GenD and at or above Effort and Forensics Adapter in the FF++ domain, above Forensics Adapter in the Celeb-DF domain, and 0.012–0.024 behind DFD-HR in both domains and GenD in the Celeb-DF domain. On reenactment it is 0.076–0.094 behind GenD and DFD-HR in the FF++ domain and 0.131–0.182 behind GenD in the Celeb-DF domain, and on par with or above Effort and Forensics Adapter except in the Celeb-DF domain on the first seed (Effort -0.063; Table S23). OURS, full model is behind every CLIP detector on reenactment, by 0.090–0.218 in the FF++ domain and 0.163–0.326 in the Celeb-DF domain, at chance in the latter (0.502); its 0.015 gap to GenD on FF++-domain face swap contains zero under identity resampling. P3, which never sees smoothed genuine faces, is the best exposure arm on Celeb-DF-domain reenactment (0.753 against 0.692 for P0); across the 13 published detectors scored on both sets the share of smoothed renders ranked above their original does not significantly predict reenactment AUC (Spearman $\rho = 0.36$ and 0.49, $p = 0.23$ and 0.09), so we do not claim a link beyond our arms.

Entire-face synthesis. A descriptive probe on six DF40 synthesis methods (300 images per method and domain; the face-editing sets listed by DF40 were not available to us) finds our models weakest where the released CLIP detectors are strongest: below Effort and GenD on every method and domain (Table S24; sd2.1 92.0 and 73.0% above the 95th percentile of same-domain real frames for OURS, forgery head only, 72.0 and 56.3% for OURS, full model, against 99.0 and 87.0% for Effort). The full model's three-way output calls 88–89% of ddim stills and 42–48% of SiT stills *beautified*: smooth

synthesised skin is absorbed by the beautified class although the definition says forged. The variant is the weakest on this probe (StyleGAN3 in the Celeb-DF domain 5.3%).

F. Localisation and restoration of part-level forgeries

For a detected part-level forgery with footprint M, let $\hat{A}$ be the $|M|$ highest cells of $\hat{m}$ and $x_A = x \odot (1 - A) + x_0 \odot A$ restore region A to the original. These results are those of the evidence head of Ours, full model (the forgery-head-only model has none); it reaches IoU_7 (IoU with the footprint on a 7×7 grid) of 0.83–0.84 on three held-out editors (SDXL never used in training), and restoring the region it names removes 93.5–94.0% of its detections against 93.6–94.1% for the true footprint (Table S31; second seed 90.2–90.9% against 90.2–91.0%). Restoring the complement removes at most 0.4%, a centred region 39–41%; post-hoc maps without the head reach IoU_7 0.33–0.41. Any accurate segmenter of the edit would score similarly, so the test does not show the classifier's reasoning, and it does not cover whole-face swaps or accused genuine faces.

G. What did not work

Under pre-registered bars (Sec. S19): adversarial or random training over a measured beautification operator (seven variants) stays within ± 0.01 AUC of colour grading and loses 0.02–0.05 untreated AUC; a learned inverse of the edit does not lower false alarms; a decision pooled from the evidence map halves false alarms under filters and lets 7.5 points more beautified forgeries through; fine-tuning released Effort with beautified views moves its operating point, not its AUC; fusion with CADDM still ranks smoothed renders above their originals in 87–92% of pairs (Sec. S18).

VI. Limitations

The measurement covers four app presets, 27 colour filters, eight AR filters on two face sets and one retouching service that has become an analysis set; a fresh smoothing source is needed, and the AR capture path is confounded with the beautification. The class boundary is stated, not measured, and DF40's attribute-editing sets were not available. P3 has one seed; P1 and P3 were not trained under Effort's scheme, and no non-CLIP pipeline was trained. The symmetric view does not cover skin-only smoothing, and its miss-rate cost differs between seeds (34.1 and 11.6 points). Our models accuse AR renders (Ours, forgery head only 79.0–90.3%), so we do not claim that our training removes the accusation in general. Ours, forgery head only is behind GenD and DFD-HR on DF40 reenactment and behind Effort and GenD on entire-face synthesis; the full model's beautified head costs detection, and its three-way output sends 20.8% of the service's originals and 13.2% of untreated Celeb-DF-B genuine frames to the beautified class (47.4% of beautified ones), so that attribute is not a separation result. Effort and DFD-HR were not reproduced at their published numbers, DFDC is not evaluated, our DFD set is 332 of its videos, and inference cost was not measured.

VII. Conclusion

Under one rule, all seventeen published detectors accuse Instagram AR renders on at least one face set and accuse skin-only smoothing, fifteen score a commercially smoothed genuine face above its own original in 89–99% of pairs, and for the four CLIP detectors the forgery logit grows with the smoothing strength. In controlled training the class that sees beautified views sets the sign of the error, and a class-symmetric smoothing view removes the accusation only for the forms it was drawn from and only under LoRA tuning. Our models accuse AR renders as much as the published CLIP detectors, and the full model's beautified head costs detection against the same model with the forgery head only; these are measurements and training lessons, not a solved problem.

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